

FoodAccessPREP

Oliver Temel, Mairangi Campbell and Jiayi Lu

2026-08-19

Background and Data

Our group used the 2015 and 2019 Food Access Research Atlas datasets, available through the U.S. Department of Agriculture Economic Research Service.

The datasets are of interest because they provide insight into access to supermarkets and essential food resources across different communities. They can be used to examine food access among people receiving benefits, those living in areas of high deprivation or low socioeconomic status, those living further away in urban vs rural areas, and different demographic groups. Using data from multiple years also allows us to identify changes over time and determine whether there have been noticeable improvements in food access.

The datasets contain information including census tract, poverty rate, food access at specified distances, and access measures across different racial and age groups, as well as SNAP beneficiaries. The majority of the variables are numerical, with some categorical variables such as census tract, state, and county. The data is structured so that individual variables represent specific measures of food access for particular demographic groups and distance thresholds.

One point of interest was that the 2019 dataset contained a large number of NULL values. These were assumed to represent zero values, as many of the affected variables represented whether an observation belonged to a particular demographic or access category. For example, a variable could represent the number of Native American individuals experiencing low food access at 20 miles, where a NULL indicated that there were no individuals meeting that condition. In contrast, both datasets also contained a considerable number of genuine NA values, particularly for `SNAP_la_rate` (2,405) and `senior_la_rate` (520).

To enable comparisons between years, our group used both the 2015 and 2019 Food Access Research Atlas datasets. The datasets were straightforward to integrate because they contained consistent and largely homogeneous variable names. We therefore retained the same variables from each dataset, allowing us to make meaningful comparisons in food access between 2015 and 2019.

Ethics, Privacy & Security

Ethics

This project involves data describing vulnerable populations, specifically seniors aged 65 and over and SNAP-receiving households. These populations require discretion as careless findings or framing about them can further stigmatise affected communities. A key consideration for this analysis is avoiding deficit oriented framing, practically, this means describing a tract as “low access” reflects a structural and geographic outcome specifically the distance for a household to supermarkets or transport infrastructure. It is not intended to signify the choices of the individuals living in the household, this is to avoid perpetuating any stereotypes or stigmas. As a result the language of the report has been mindfully selected to reflect this. For example we would state “A tract has a high rate of low-access seniors” instead of language that could be read as describing a senior or snap receiving households like “seniors in this area don’t shop at super markets”

Another ethical risk we are considering is an ecological fallacy. All analysis in this project is conducted at the census-tract level which uses aggregated rates for an area rather than individual data. So where a tract is

classified as “low access” indicates the aggregate population meets USDA distance thresholds, it does not mean that every individual within that tract lacks access. As a result our conclusions are based on patterns observed in tracts, not on individuals.

Privacy

This project uses only publicly available, government-released data (USDA Food Access Research Atlas and U.S. Census Bureau data), both provided at the census-tract level. We did no original data collection of our own, or have any interaction with the individuals involved with the data. The data is already aggregated tract-areas and released for public use as verified by the U.S. government so the privacy risk to individuals is minimal. The risks too to privacy are not technical with how the data is structured or analysed but in the interpretation of the data. There is still a risk (although very small risk) of individual identification for tracts with very small populations e.g a small amount of senior SNAP households in a rural tract, where in principle it could be easier to infer information about specific households. This represents a limitation of the dataset from a privacy perspective, though it is a low and indirect risk rather than a risk introduced by our analysis.

Security

As mentioned in the privacy section this project uses only public, pre-aggregated data. The overall security risk associated with this project is low. We practiced secure data management in this project where we store all our working data and code within a shared private Github repository, accessible only to the members of our group as opposed to a public repository or sharing code in any other external source. This gave us a controlled and access-limited location for our data, code and analysis. With some additional time and resources to allocate into security it would be worthwhile establishing a documented data deletion plan for after the project concludes.

After harmonisation, the 2015 and 2019 datasets were combined into a dataset containing 145,395 census-tract observations. Of these, 110,341 were classified as urban and 35,054 as rural, with no missing values in the urban–rural classification.

Exploratory Data Analysis

3.1 Overview of the Analysis

The exploratory analysis focused on the low-supermarket-access shares for adults aged 65 years and over and SNAP-receiving households. Comparisons were made between 2015 and 2019 and between urban and rural census tracts. For urban census tracts, low supermarket access was measured using the one-mile threshold, while the ten-mile threshold was used for rural census tracts. All low-access shares were expressed as percentages to ensure consistency between the 2015 and 2019 datasets.

Table 1 summarises the mean and median low-access shares by year and urban/rural status. Urban census tracts generally had higher low-access shares than rural census tracts for both seniors and SNAP households. In addition, the mean was consistently higher than the median, suggesting that the distributions were right-skewed. More detailed distributions are examined in the following sections.

```
summary_table <- combined_data %>%
  group_by(area, year) %>%
  summarise(
    Senior_Mean = round(mean(senior_la_pct, na.rm = TRUE), 1),
    Senior_Median = round(median(senior_la_pct, na.rm = TRUE), 1),
    SNAP_Mean = round(mean(snap_la_pct, na.rm = TRUE), 1),
    SNAP_Median = round(median(snap_la_pct, na.rm = TRUE), 1),
    n_tracts = n(),
    .groups = "drop"
  )

kable(summary_table,
```

```
caption = "Mean and median low-access share (%) for seniors and SNAP households, by area and year"
col.names = c("Area", "Year", "Senior Mean", "Senior Median",
              "SNAP Mean", "SNAP Median", "N Tracts"))
```

Table 1: Mean and median low-access share (%) for seniors and SNAP households, by area and year.

Area	Year	Senior Mean	Senior Median	SNAP Mean	SNAP Median	N Tracts
Rural	2015	1.7	0.0	1.3	0.0	17692
Rural	2019	4.0	1.7	2.8	1.0	17362
Urban	2015	3.4	0.6	2.9	0.3	55172
Urban	2019	5.3	3.5	4.3	1.8	55169

Table 2 shows substantial differences in data completeness between the two years. There were no missing values in the Senior or SNAP low-access share variables used for 2015, whereas the 2019 data contained 29,953 missing senior observations and 29,918 missing SNAP observations. Therefore, comparisons involving 2019 should be interpreted with caution because they are based only on census tracts with available low-access information.

```
missing_table <- data.frame(
  Year = c(2015, 2019),
  Senior_Missing = c(
    sum(is.na(data2015_clean$senior_la_pct)),
    sum(is.na(data2019_clean$senior_la_pct))
  ),
  SNAP_Missing = c(
    sum(is.na(data2015_clean$snap_la_pct)),
    sum(is.na(data2019_clean$snap_la_pct))
  )
)

kable(missing_table,
      caption = "Number of missing Senior and SNAP low-access share observations, by year.",
      col.names = c("Year", "Senior Missing", "SNAP Missing"))
```

Table 2: Number of missing Senior and SNAP low-access share observations, by year.

Year	Senior Missing	SNAP Missing
2015	0	0
2019	29953	29918

3.2 Senior Low-Access Share: Distribution and Urban–Rural Comparison

Figure 1 shows the distribution of senior low-access shares by year and urban/rural status. The distributions were strongly right-skewed across all four groups, with most census tracts having relatively low senior low-access shares and a smaller number having much higher values. Urban census tracts generally showed higher low-access shares than rural census tracts. The 2019 distributions also appeared shifted toward higher values than in 2015. However, interpret the 2019 results cautiously because the senior low-access variable contained substantial missing data.

```
ggplot(combined_data, aes(x = senior_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  facet_grid(area ~ year) +
  labs(
    title = "Figure 1: Distribution of Senior Low-Access Share",
    x = "Senior Low-Access Share (%)",
    y = "Number of Census Tracts"
  ) +
  theme_minimal()
```

Figure 1: Distribution of Senior Low-Access Share

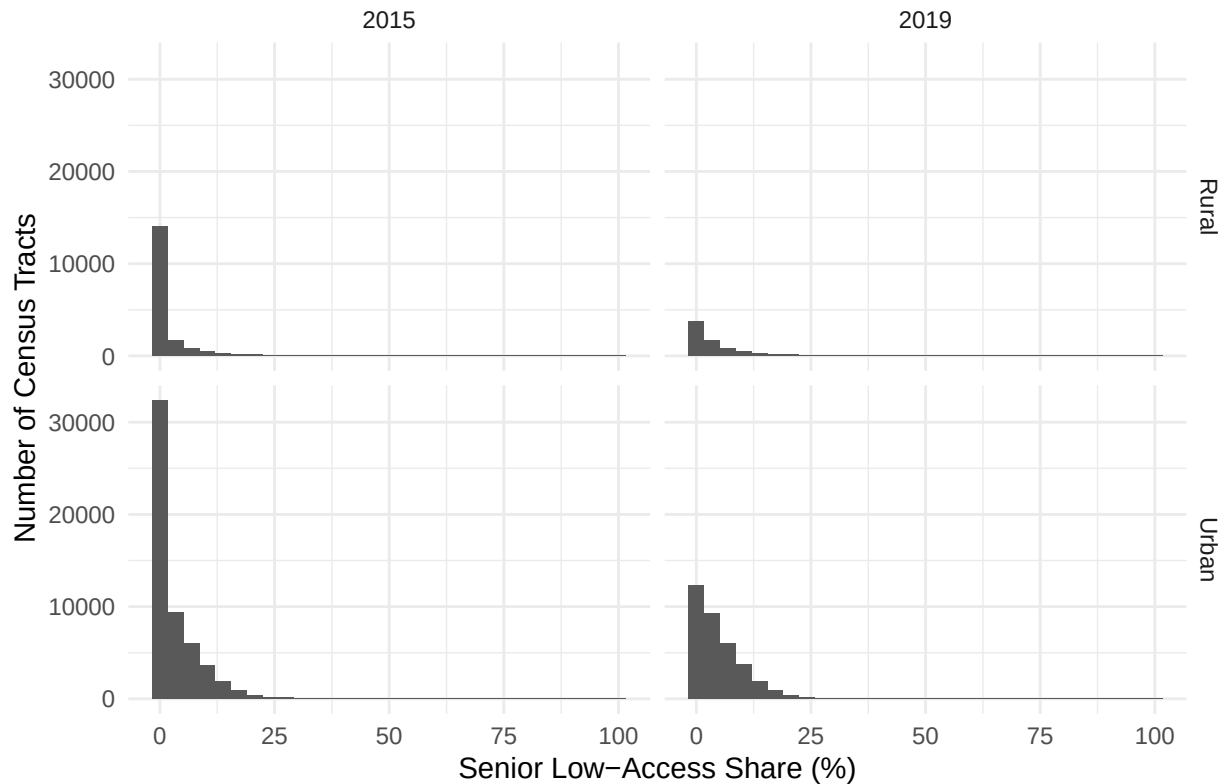


Figure 2 provides a more direct comparison between urban and rural census tracts. Urban tracts had higher median senior low-access shares than rural tracts in both 2015 and 2019, and the 2019 distributions were generally higher than those in 2015. Both years also contained several high-value outliers, which is consistent with the right-skewed distributions observed in Figure 1.

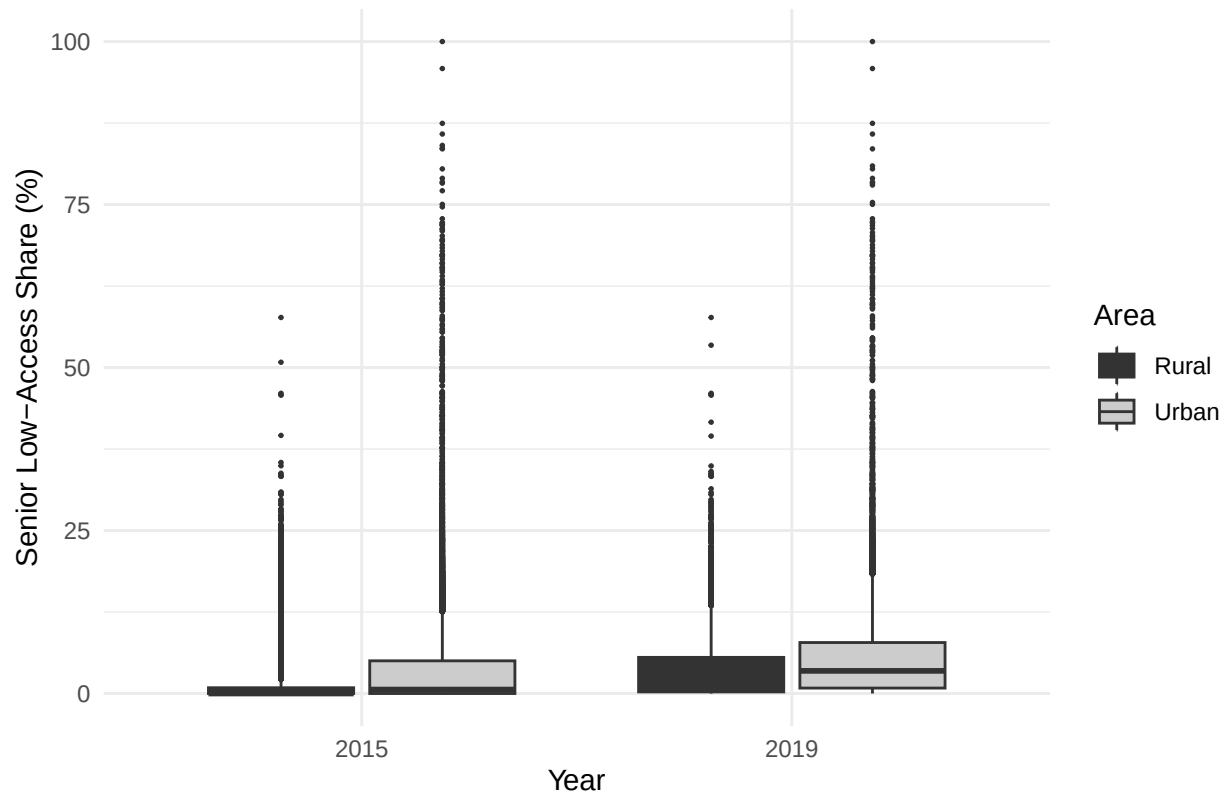
```
# Senior Boxplot
ggplot(
  combined_data,
  aes(
    x = factor(year),
    y = senior_la_pct,
    fill = area
  )
) +
  geom_boxplot(
    outlier.size = 0.3,
```

```

na.rm = TRUE
) +
scale_fill_grey() +
labs(
  title = "Figure 2: Senior Low-Access Share by Year and Area",
  x = "Year",
  y = "Senior Low-Access Share (%)",
  fill = "Area"
) +
theme_minimal()

```

Figure 2: Senior Low-Access Share by Year and Area



Overall, the senior low-access share was generally higher in urban than rural census tracts, and the distributions were strongly right-skewed in both years. The apparent differences between 2015 and 2019 should be interpreted cautiously because of substantial missing data in 2019.

3.3 SNAP Low-Access Share: Distribution and Urban–Rural Comparison

Figure 3 shows the distribution of SNAP low-access shares across census tracts. As with the senior measure, the distribution is right-skewed, with many census tracts concentrated at relatively low percentages and a smaller number of tracts having much higher values.

```

ggplot(combined_data, aes(x = snap_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Figure 3: Distribution of SNAP Low-Access Share, All Years and Areas",
    x = "SNAP Low-Access Share (%)",
    y = "Number of Census Tracts"
  )

```

```
) +  
theme_minimal()
```

Figure 3: Distribution of SNAP Low-Access Share, All Years and Areas

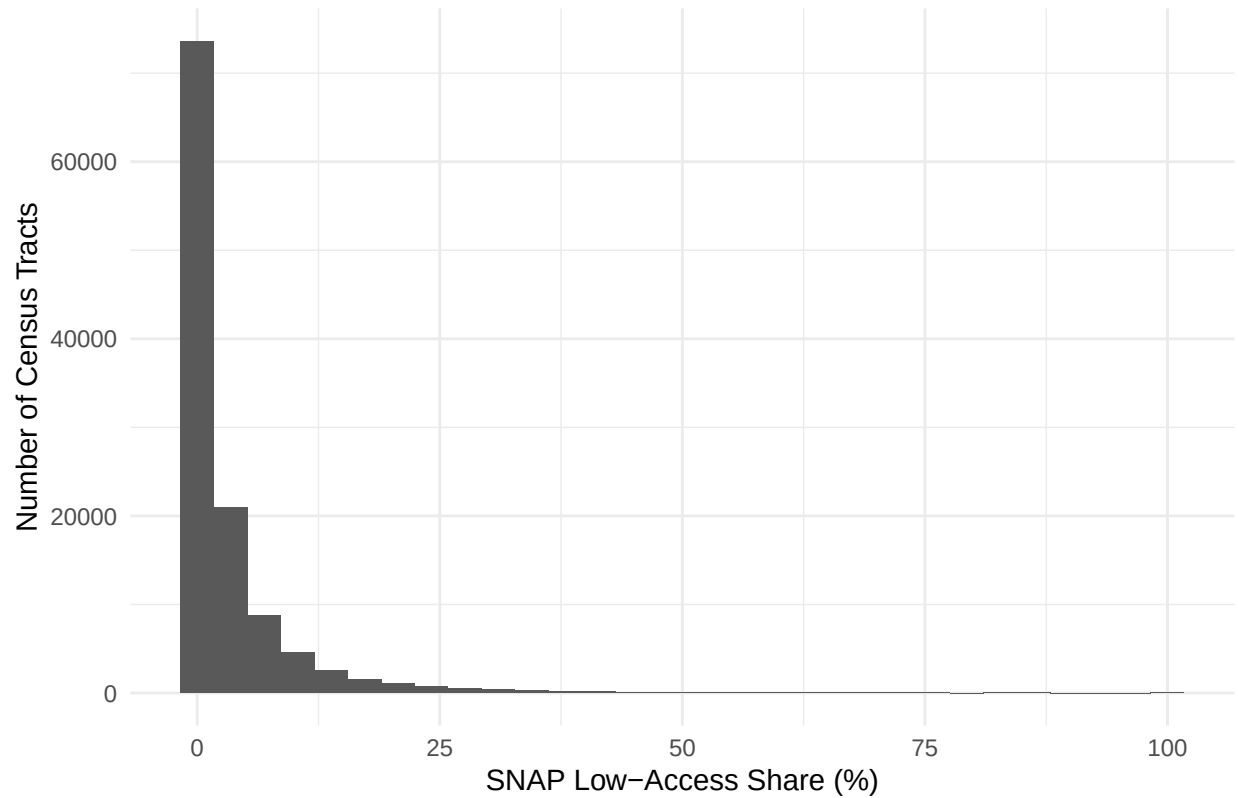
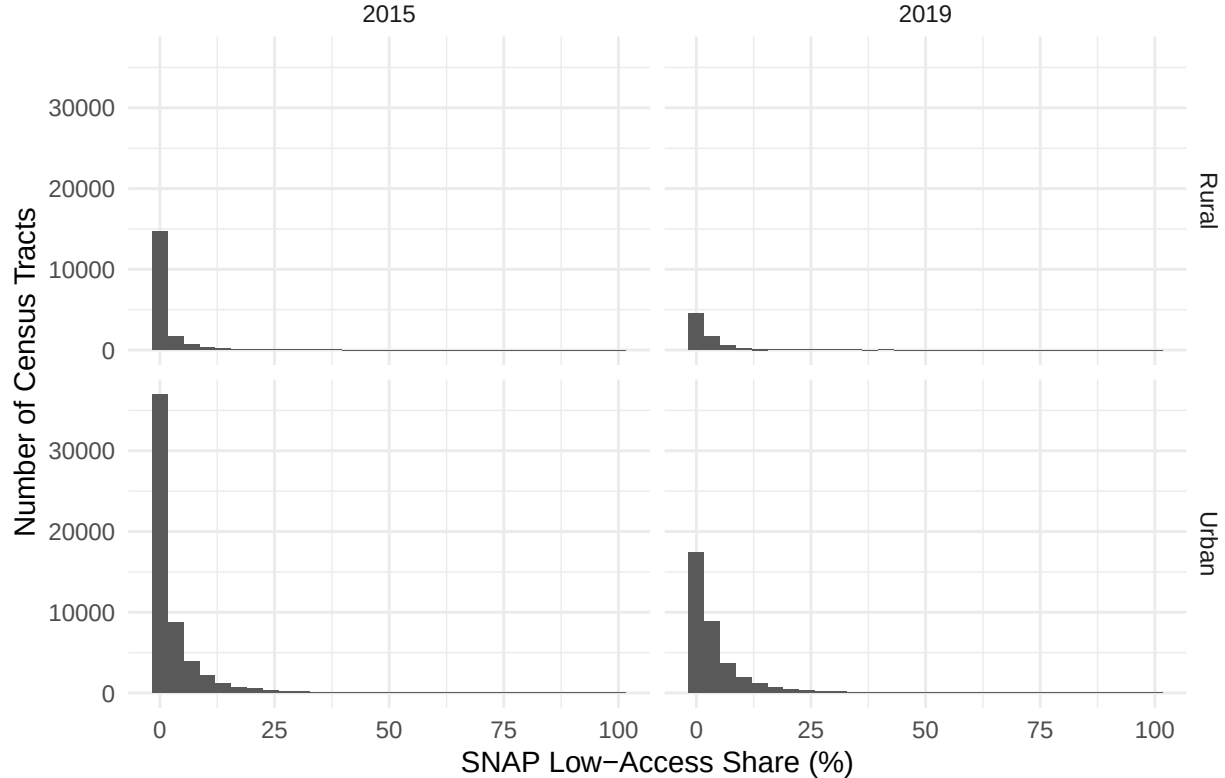


Figure 4 compares the distribution of SNAP low-access shares between urban and rural census tracts in 2015 and 2019. The plot allows differences in median, spread and extreme values to be compared while also showing whether the urban-rural pattern remained similar across the two years.

```
ggplot(combined_data,aes(x = snap_la_pct)) +  
  geom_histogram(bins = 30, na.rm = TRUE) +  
  facet_grid(area ~ year) +  
  labs(  
    title = "Figure 4: Distribution of SNAP Low-Access Share",  
    x = "SNAP Low-Access Share (%)",  
    y = "Number of Census Tracts"  
  ) +  
  theme_minimal()
```

Figure 4: Distribution of SNAP Low-Access Share



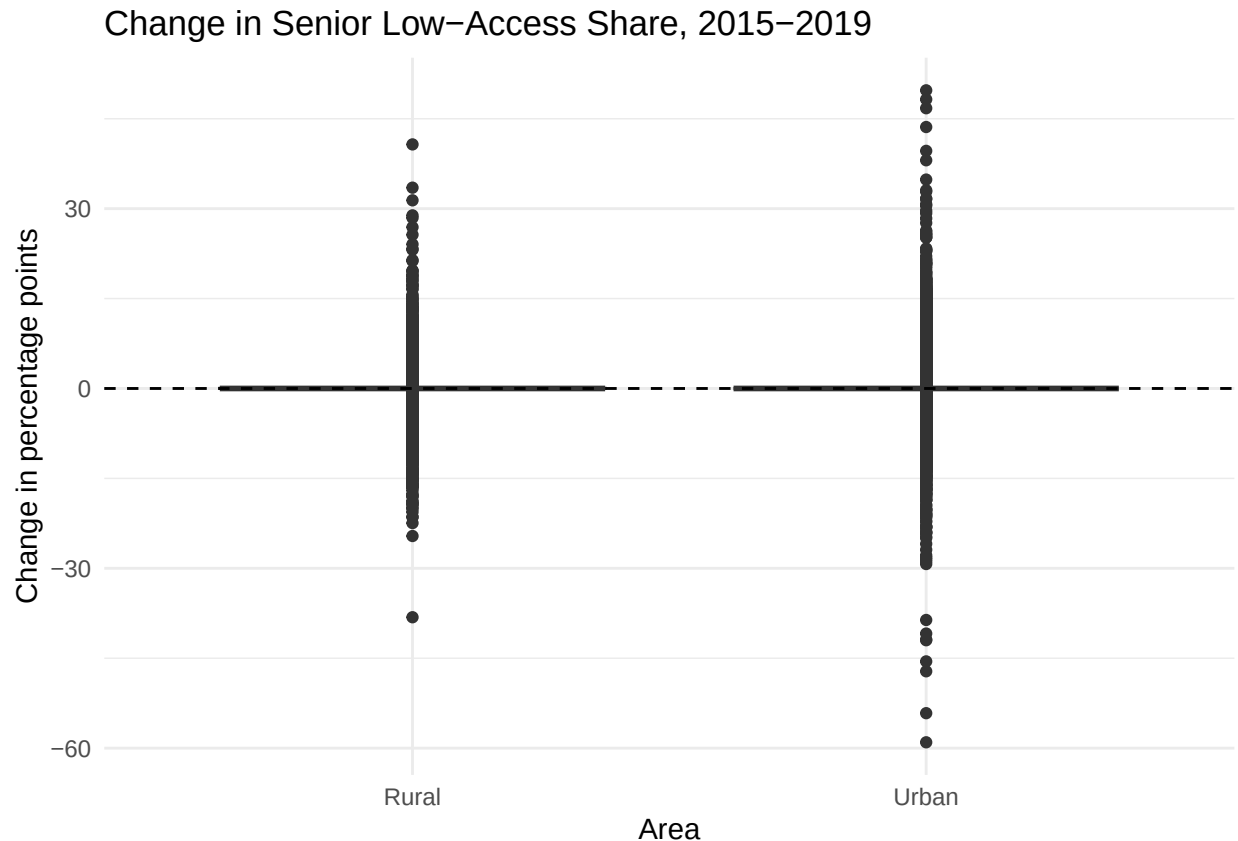
Overall, SNAP low-access shares were generally higher in urban than rural census tracts, with strongly right-skewed distributions in both years. The apparent differences between 2015 and 2019 should be interpreted cautiously because of substantial missing data in 2019.

3.4 Changes between 2015 and 2019

To examine changes between 2015 and 2019 more directly, census tracts in the two datasets were matched using their Census Tract identifiers.

A total of 72,506 census tracts were successfully matched across the two years. Of these, 21 census tracts changed their urban/rural classification between 2015 and 2019. These tracts were excluded from the change analysis because the low-access distance threshold differs between urban and rural areas. This resulted in 72,485 census tracts with a stable urban/rural classification.

```
# Senior Change (using all matched tracts, before excluding reclassified tracts)
ggplot(
  matched_data,
  aes(x = area_2015, y = senior_change)) +
  geom_boxplot(na.rm = TRUE) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Change in Senior Low-Access Share, 2015-2019",
    x = "Area",
    y = "Change in percentage points"
  ) +
  theme_minimal()
```



Reason for why choose histogram to show the Change in Senior Low-Access Share between 2015 and 2019: Based on the summary, we find the boxplot is right. However, the vast majority of changes in census tracts are very close to zero. From the summary, especially Urban, $Q1 = -4.019 \times 10^{-14}$ Median = 0 $Q3 = 4.974 \times 10^{-14}$ The real meaning of these three number is 0% points. Thus, the boxplot of Urban: $Q1 \approx \text{Median} \approx Q3 \approx 0$, it shows the box was naturally compressed into a line near $y=0$.

While, Rural is not completely 0: $Q1 \approx -0.014$ Median = 0 $Q3 \approx 0.052$ However, the IQR just around $0.052 - (-0.014) \approx 0.066$. There is the very small percentage points. Compare the Min(-38.16) and Max(+40.72), so the vertical axis is stretched very wide to accommodate these extreme changes, making the box even less visible.

The key finding We cannot be fully trust “pct_exact_zero”. You now obtain: Rural: 10.41% Urban: 15.50% It seems that only 10–15% have remained unchanged.

However, for Urban, $Q1$ and $Q3$ are: -0.0000000000000040 $+0.0000000000000050$

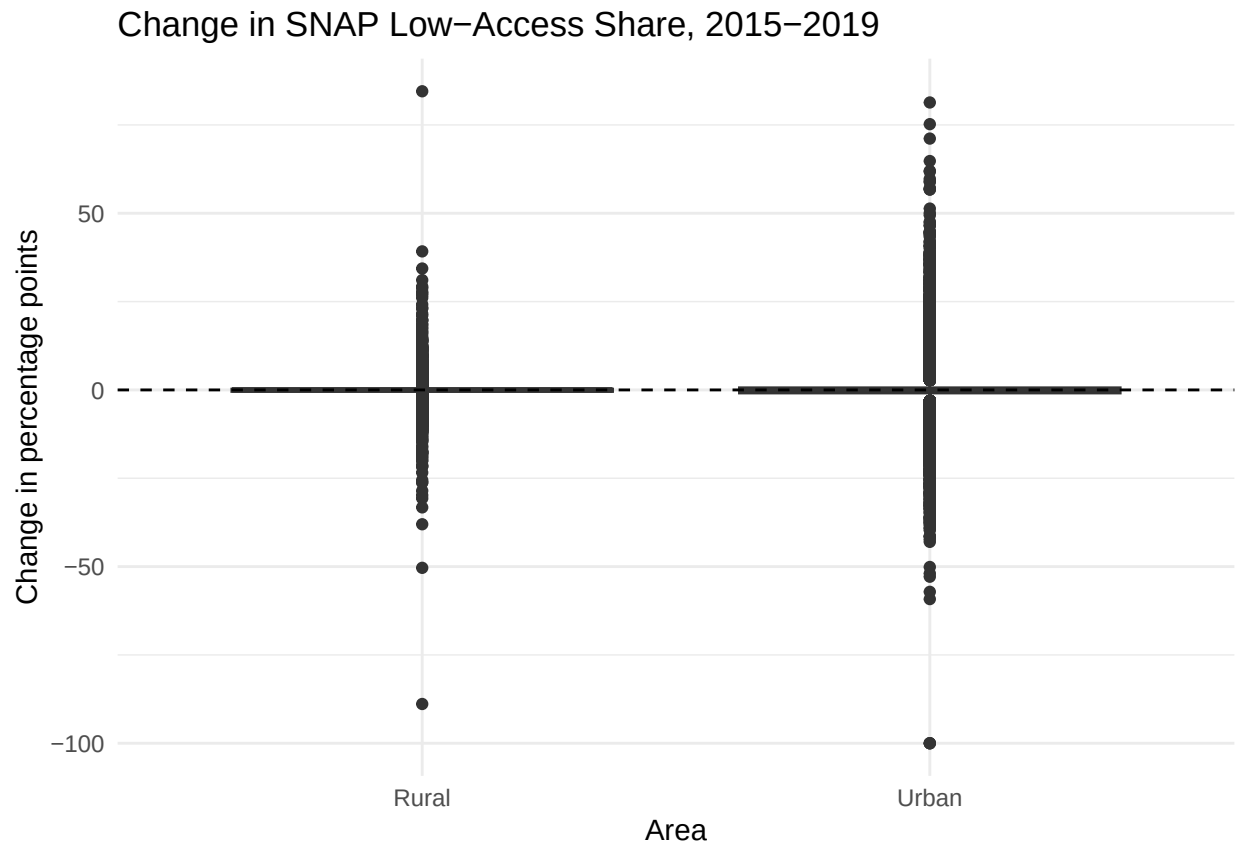
This is actually floating-point numerical noise. R will consider: $0.000000000000005 \neq 0$ So you use: `senior_change == 0`

It will classify many observations that are actually equivalent to 0 as “non-zero changes”.

```
# SNAP
ggplot(
  matched_data,
  aes(x = area_2015, y = snap_change)) +
  geom_boxplot(na.rm = TRUE) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Change in SNAP Low-Access Share, 2015-2019",
```



```
x = "Area",
y = "Change in percentage points"
) +
theme_minimal()
```



```
matched_stable %>%
  group_by(area_2015) %>%
  summarise(
    n = sum(!is.na(snap_change)),
    Q1 = quantile(snap_change, 0.25, na.rm = TRUE),
    Median = median(snap_change, na.rm = TRUE),
    Q3 = quantile(snap_change, 0.75, na.rm = TRUE),
    Min = min(snap_change, na.rm = TRUE),
    Max = max(snap_change, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 7
##   area_2015      n      Q1   Median    Q3     Min    Max
##   <chr>      <int> <dbl>   <dbl> <dbl> <dbl> <dbl>
## 1 Rural       7399 -0.434 -0.00152 0.232  -88.9   84.5
## 2 Urban      35190 -0.866  0        0.603 -100.   81.4
```

```
matched_stable %>%
  group_by(area_2019) %>%
  summarise(
    n = sum(!is.na(snap_change)),
```

```

Q1 = quantile(snap_change, 0.25, na.rm = TRUE),
Median = median(snap_change, na.rm = TRUE),
Q3 = quantile(snap_change, 0.75, na.rm = TRUE),
Min = min(snap_change, na.rm = TRUE),
Max = max(snap_change, na.rm = TRUE),
.groups = "drop"
)

```

```

## # A tibble: 2 x 7
##   area_2019      n    Q1  Median    Q3    Min    Max
##   <chr>    <int> <dbl>   <dbl> <dbl> <dbl> <dbl>
## 1 Rural      7399 -0.434 -0.00152 0.232  -88.9   84.5
## 2 Urban     35190 -0.866  0        0.603 -100.   81.4

```

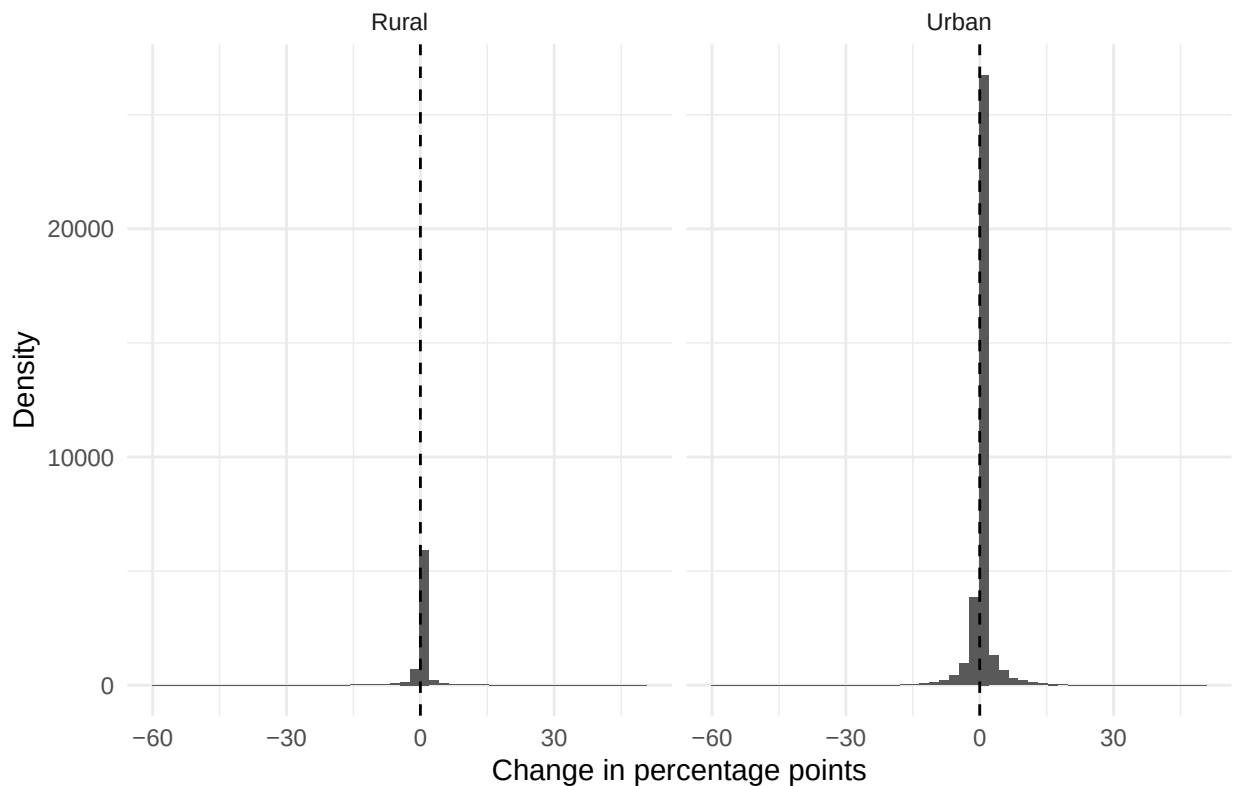
Still check, Q1, Median, and Q3 all close to 0, it shows the data itself is highly concentrated around 0.

```

ggplot(matched_data, aes(x = senior_change)) +
  geom_histogram(bins = 50, na.rm = TRUE) +
  facet_wrap(~ area_2015) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Distribution of Changes in Senior Low-Access Share, 2015-2019 (unfiltered)",
    x = "Change in percentage points",
    y = "Density"
  ) +
  theme_minimal()

```

Distribution of Changes in Senior Low-Access Share, 2015-2019 (unfiltered)



This will show more clearly than boxplot: A large number of tracts are concentrated around 0; Which ones are increases? Which ones are reductions? Is the distribution of Urban and Rural areas different?

We should use the second which better than last one. Even though we cannot see observable difference, it shows the results which checked: Rural to Urban: 10; Urban to Rural: 11 . There are 21 census tracts deleted.

Therefore, although the second graph does not show any visual difference, it is more rigorous in its approach because it ensures that: For rural areas, the comparison always uses a 10-mile measure, and for urban areas, the comparison always uses a 1-mile measure. So, the subsequent 2015 - 2019 change analysis suggests using "matched_stable".

For the paired analysis, only census tracts with non-missing low-access shares in both 2015 and 2019 were included. Change was calculated as: Change= Low-access share2019 - Low-access share2015 and is reported in percentage points. Positive values therefore represent an increase in low-access share between 2015 and 2019, while negative values represent a decrease.

```
paired_summary <- bind_rows(  
  senior_paired %>%  
    mutate(Group = "Senior"),  
  
  snap_paired %>%  
    mutate(Group = "SNAP")  
) %>%  
  select(  
    Group,  
    area_2015,  
    N,  
    Mean_2015,  
    Mean_2019,  
    Median_2015,  
    Median_2019,  
    Mean_Change,  
    Median_Change  
  ) %>%  
  arrange(Group, area_2015)  
  
kable(  
  paired_summary,  
  caption = "Paired summary of Senior and SNAP low-access shares, 2015-2019",  
  col.names = c(  
    "Group",  
    "Area",  
    "N",  
    "Mean 2015 (%)",  
    "Mean 2019 (%)",  
    "Median 2015 (%)",  
    "Median 2019 (%)",  
    "Mean Change (pp)",  
    "Median Change (pp)"  
  ),  
  align = c(  
    "l", "l", "r",  
    "r", "r", "r", "r", "r", "r"  
  )  
)
```

Table 3: Paired summary of Senior and SNAP low-access shares, 2015-2019

Group	Area	N	Mean 2015 (%)	Mean 2019 (%)	Median 2015 (%)	Median 2019 (%)	Mean Change (pp)	Median Change (pp)
SNAP	Rural	7399	2.91	2.77	1.00	0.96	-0.14	0
SNAP	Urban	35190	4.40	4.33	1.76	1.77	-0.07	0
Senior	Rural	7387	3.94	3.96	1.56	1.72	0.02	0
Senior	Urban	35167	5.21	5.32	3.35	3.47	0.11	0

From Table 3, we compared the different groups with the different areas between 2015 and 2019. For seniors, the paired mean low-access share increased only slightly between 2015 and 2019. In rural census tracts, the mean increased from 3.94% to 3.96%, corresponding to an average increase of 0.02 percentage points. In urban census tracts, the mean increased from 5.21% to 5.32%, an average increase of 0.11 percentage points. The median change was zero in both rural and urban tracts, indicating that the typical matched census tract experienced little or no change. For SNAP households, the direction of the average change differed from that observed for seniors. The mean low-access share decreased from 2.91% to 2.77% in rural census tracts, corresponding to a mean change of -0.14 percentage points. In urban census tracts, the mean decreased from 4.40% to 4.33%, corresponding to a mean change of -0.07 percentage points. However, the median change was again zero in both groups, suggesting that changes for a typical matched census tract were very small.

Figure 5 shows that changes in senior low-access share were strongly concentrated around zero in both rural and urban census tracts. This is consistent with the zero median changes reported in Table 3. Although most census tracts experienced relatively small changes, a smaller number showed substantially larger positive or negative changes.

```
ggplot(matched_stable, aes(x = senior_change)) +
  geom_histogram(bins = 50, na.rm = TRUE) +
  facet_wrap(~ area_2015) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Figure 5: Distribution of Changes in Senior Low-Access Share, 2015-2019",
    x = "Change in percentage points",
    y = "Density"
  ) +
  theme_minimal()
```

Figure 5: Distribution of Changes in Senior Low-Access Share, 2015–20.

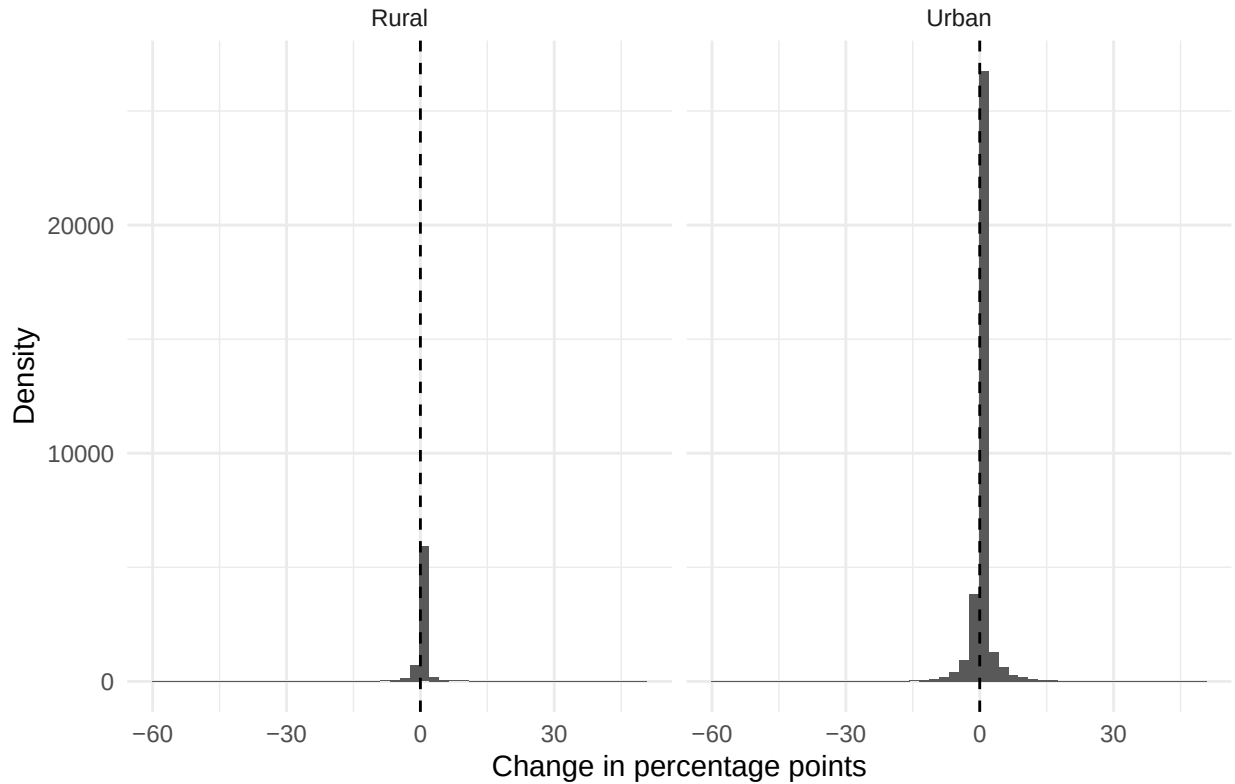
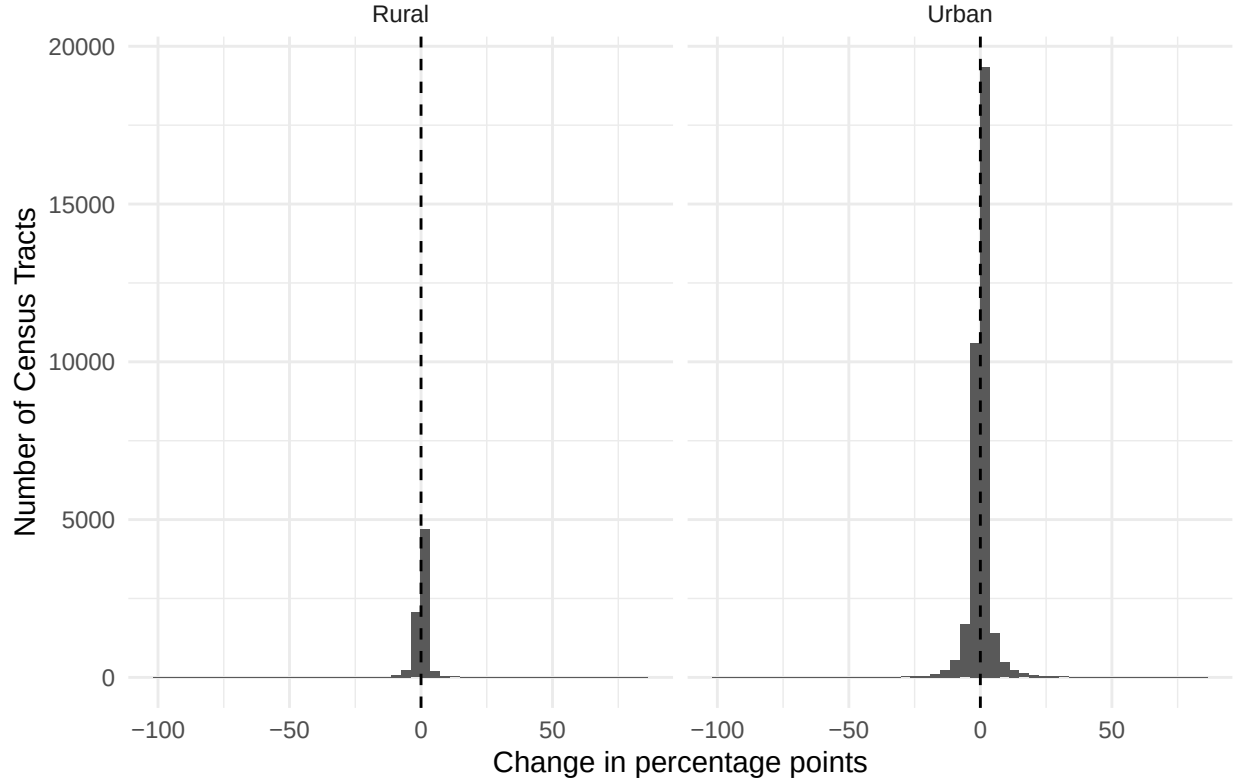


Figure 6 shows that change distributions in SNAP Low-Access share were also strongly concentrated around zero, although some census tracts experienced larger increases or decreases. The concentration around zero is consistent with the paired summary statistics and indicates that the overall changes between the two years were small for most matched census tracts.

```
ggplot(matched_stable, aes(x = snap_change)) +
  geom_histogram(bins = 50, na.rm = TRUE) +
  facet_wrap(~ area_2015) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Figure 6: Distribution of Changes in SNAP Low-Access Share, 2015-2019",
    x = "Change in percentage points",
    y = "Number of Census Tracts"
  ) +
  theme_minimal()
```

Figure 6: Distribution of Changes in SNAP Low-Access Share, 2015–2019



```
matched_stable %>%
  group_by(area_2015) %>%
  summarise(
    n = sum(!is.na(snap_change)),
    Q1 = quantile(snap_change,0.25,na.rm = TRUE),
    Median = median(snap_change,na.rm = TRUE),
    Q3 = quantile(snap_change,0.75,na.rm = TRUE),
    Min = min(snap_change,na.rm = TRUE),
    Max = max(snap_change,na.rm = TRUE),
    pct_near_zero =mean(abs(snap_change) < 1e-8,na.rm = TRUE) * 100,
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 8
##   area_2015      n      Q1   Median    Q3    Min    Max pct_near_zero
##   <chr>      <int> <dbl>   <dbl> <dbl> <dbl> <dbl>      <dbl>
## 1 Rural      7399 -0.434 -0.00152 0.232 -88.9  84.5         3.32
## 2 Urban     35190 -0.866  0        0.603 -100.  81.4         4.25
```

Overall, the paired analysis suggests that tract-level changes between 2015 and 2019 were generally small. Senior low-access shares showed slight positive average changes in both rural and urban areas, whereas SNAP low-access shares showed slight negative average changes. However, median changes were zero for all four comparisons, indicating that the mean changes were influenced by a smaller number of census tracts with larger positive or negative changes. These findings are descriptive and do not by themselves establish whether the observed differences are statistically significant.

3.5 Additional Comparison: Mean Low-Access Share by Group

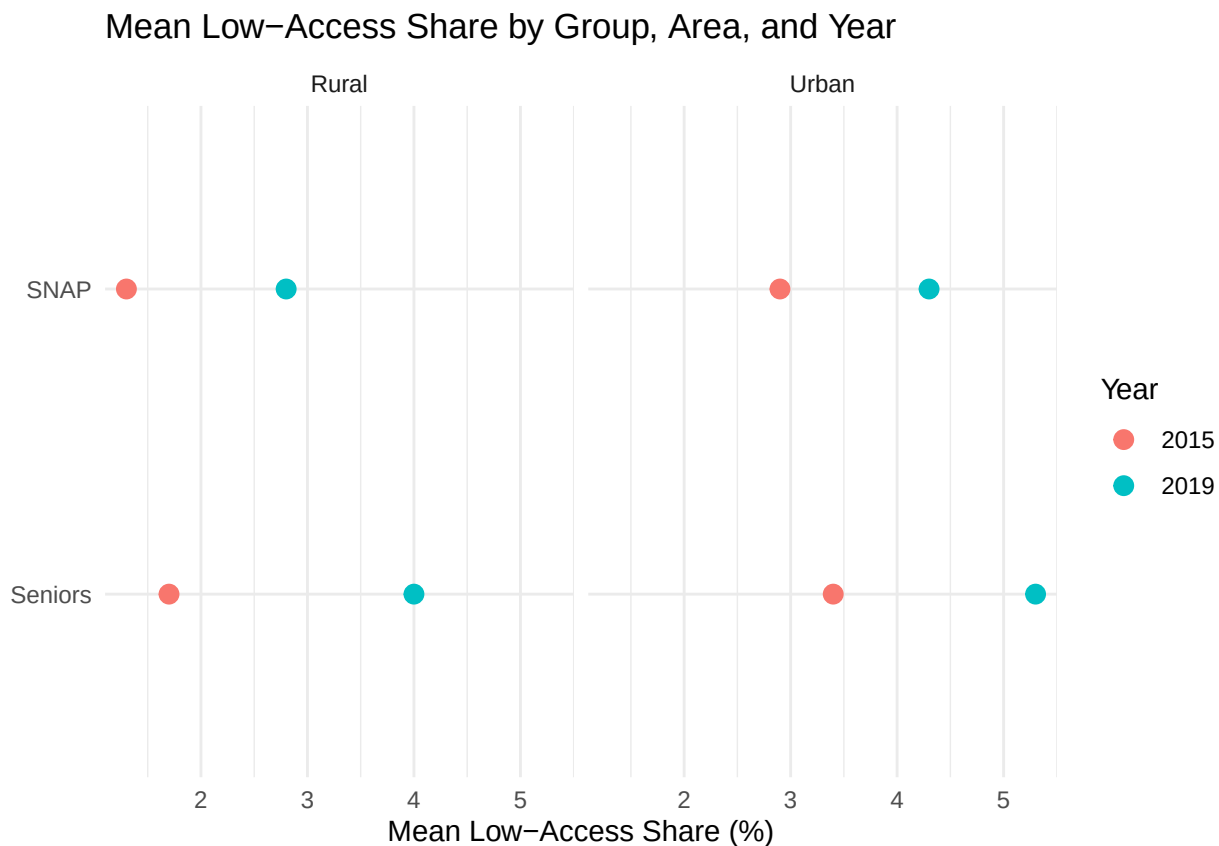
To talk about... median much lower than means = skewed data. Urban vastly outnumber rural. Values increase across all four rows

keeping in mind a rural tract requires households and seniors to be at least 10 miles away compared to a min of 1

```
library(tidyr)

dot_data <- summary_table %>%
  select(area, year, Senior_Mean, SNAP_Mean) %>%
  pivot_longer(cols = c(Senior_Mean, SNAP_Mean),
               names_to = "Group", values_to = "Mean_Rate") %>%
  mutate(Group = recode(Group, Senior_Mean = "Seniors", SNAP_Mean = "SNAP"))

ggplot(dot_data, aes(x = Mean_Rate, y = Group, color = factor(year))) +
  geom_point(size = 3) +
  facet_wrap(~area) +
  labs(
    title = "Mean Low-Access Share by Group, Area, and Year",
    x = "Mean Low-Access Share (%)",
    y = NULL,
    color = "Year"
  ) +
  theme_minimal()
```



Across both Senior and SNAP low-access shares, 2019 shows a higher mean rate than 2015, and Urban tracts

show a higher mean rate than Rural tracts, in both years and for both groups

Because “low-access” is defined using a much stricter distance threshold in urban areas (1 mile) than rural areas (10 miles), a higher urban low-access rate does not necessarily mean urban residents travel further to reach a supermarket than rural residents. This may just reflect that urban areas are held to a stricter access standard. A rural resident traveling several miles to a store may still count as “having access” under the rural threshold, while a much shorter distance in a city would count as “low access.”

As a result, the lower mean low-access share in rural areas should not be read as evidence that rural food access is genuinely better. Rural residents may need to travel considerably farther in absolute terms before being classified as low-access at all, meaning the rural rate likely understates the true burden of distance compared to the urban rate. This distinction should be kept in mind whenever urban and rural low-access rates are compared directly in this report, rather than treating them as measuring the same underlying concept.

3.6 Summary

The exploratory analysis identified several important patterns in the senior and SNAP low-access measures. First, both low-access share distributions were strongly right-skewed. Most census tracts had relatively low low-access shares, while a smaller number of tracts had substantially higher values. As a result, the mean was generally higher than the median, indicating that the mean was influenced by census tracts with relatively large low-access shares.

Although the 2019 cross-sectional distributions appeared higher than those in 2015, these comparisons should be interpreted cautiously because the 2019 data contained substantial missing values. The paired analysis provided a more direct comparison of the same census tracts across years and showed that changes were generally small. Senior low-access shares increased slightly on average, while SNAP low-access shares decreased slightly, and the median change was zero across both rural and urban tracts.